

HAT-P-36b

R-1416

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_210508 · created 2026-08-01T05:46:21+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459342.8102 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.12 +/- 0.01
Transit depth (Rp/Rs)^2	1.43 +/- 0.25 [%]
Orbital Inclination (inc)	86.6 +/- 2.04
Airmass coefficient 1 (a1)	1.085 +/- 0.012
Airmass coefficient 2 (a2)	-0.0604 +/- 0.007
Scatter in the residuals of the lightcurve fit is	1.07 %
Best Comparison Star	#2 - [103, 60]
Optimal Aperture	3.76
Optimal Annulus	7.14
Transit Duration (day)	0.0938 +/- 0.0037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.12 $\pm$ 0.01 vs 0.126 (deviation 0.6 σ)
Duration fit vs published	0.0938 d vs 0.0925 d (deviation 0.35 σ)
Mid-time O-C	-1.0 min
Depth SNR	12.0
Residual scatter	1.07 %

Light curve

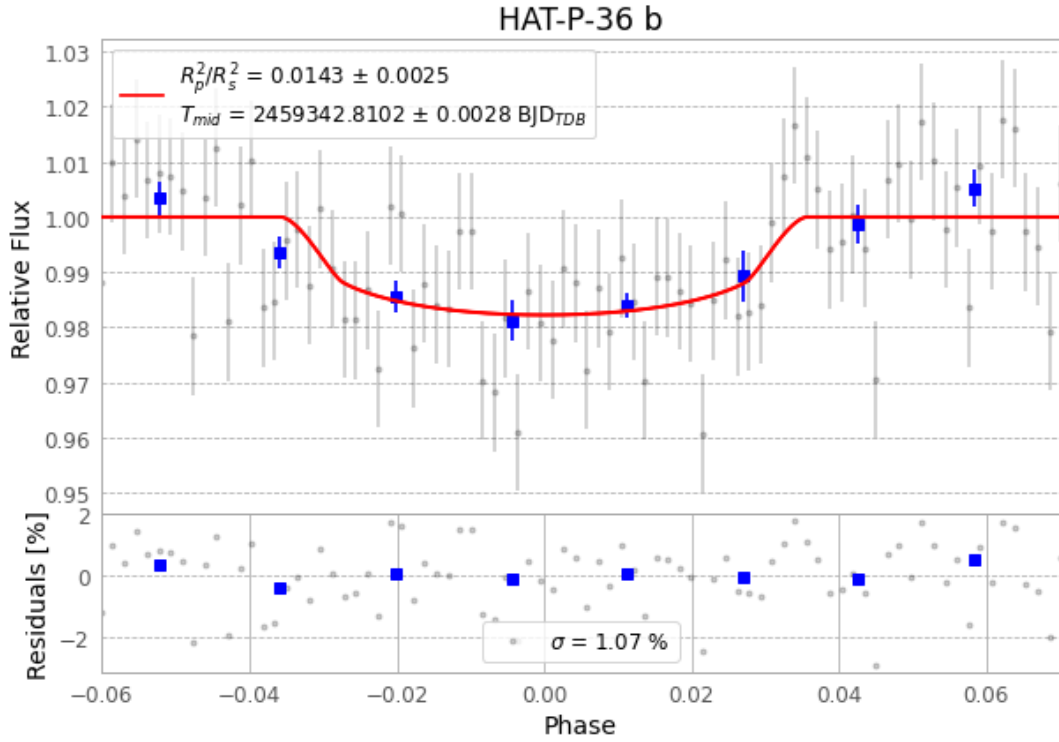


Figure 1 — FinalLightCurve_HAT-P-36 b_2021-05-07.png (SHA-256 07891530054cba09...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [407, 224], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 de6b68d790373665...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

84 FITS frames (55 MB), HATP-36210508052712.FITS → HATP-36210508093612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-210508.log (91d1645c7f81...), FinalLightCurve_HAT-P-36 b_2021-05-07.png (07891530054c...), FinalLightCurve_HAT-P-36 b_2021-05-07.csv (021157230456...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 05:46Z.